

Computer Networks: Comprehensive Lecture Notes

Unit 1: Introduction, Architectures, Services, and Reference Models

Course: Computer Networks & Security

Topic: Fundamental Concepts & Models

Level: Undergraduate / Core CS

■ 1. Introduction to Computer Networks

A **Computer Network** is a collection of autonomous computing devices interconnected by communication channels that allow them to exchange data, share hardware and software resources, and collaborate. Devices on a network—often referred to as *nodes*—can include desktop computers, servers, smartphones, routers, switches, and IoT sensors.

Uses of Computer Networks

Computer networks have fundamentally transformed modern computing across various domains:

- **Resource Sharing:** Physical resources such as printers, scanners, and storage arrays (NAS/SAN), as well as digital resources like databases and specialized software, can be shared among multiple users to optimize costs and operational efficiency.
- **High Reliability & Redundancy:** Critical files and applications can be replicated across multiple machines. If one node fails, alternate machines take over, ensuring continuous system availability.
- **Cost Reduction:** Small workstation computers linked via a server network offer a significantly higher performance-to-cost ratio compared to isolated mainframe systems.
- **Communication & Collaboration:** Networks enable instant global communication through email, IP telephony, video conferencing, real-time messaging, and collaborative workspaces.
- **Scalability & Elasticity:** System processing capacity can be increased incrementally by attaching additional client and server nodes as enterprise demands scale.

■ 2. Classification & Types of Computer Networks

Networks are primarily classified based on their **geographical scale**, physical transmission media, administrative control, and topology.

Personal Area Network (PAN)

Scale: Within a radius of ~10 meters (1 person / single room).

Centered around an individual user. Connects personal devices like smartphones, smartwatches, tablets, and wireless headphones. Uses technologies like Bluetooth, Zigbee, Ultra-Wideband (UWB), and USB.

Local Area Network (LAN)

Scale: Single room, home, office building, or campus (~10m to 2km).

Privately owned network connecting computing devices. Characterized by high data transfer speeds (1 Gbps – 100 Gbps), low latency, and low error rates. Commonly uses Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11).

Metropolitan Area Network (MAN)

Scale: Entire city or large municipality (~5km to 50km).

Spans a broader geographic area than a LAN. Often used to connect multiple corporate offices across a city or power municipal public networks (e.g., Cable TV networks, Metro Wi-Fi). Technologies include Fiber Optics, WiMAX, and Gigabit Ethernet switches.

Wide Area Network (WAN)

Scale: Country, continent, or entire globe (> 50km).

Spans vast geographical distances, connecting multiple LANs and MANs. Utilizes communication links provided by telecom common carriers (leased lines, satellite links, fiber optics). Characterized by lower speeds and higher latency compared to LANs.

Internetworks

An **Internetwork** (or simply *Internet* when referring to the global network) is created when two or more distinct, heterogeneous networks (LANs, MANs, WANs) are interconnected using hardware devices called **routers** or **gateways**. An internetwork abstracts underlying hardware differences, allowing machines on different physical networks to communicate seamlessly using a unified protocol suite (such as TCP/IP).

■ 3. Evolution of Networking: ARPANET, NSFNET, & Internet Architecture

ARPANET (Advanced Research Projects Agency Network)

- **Origins:** Commissioned in the late 1960s (1969) by the U.S. Department of Defense's Advanced Research Projects Agency (DARPA).
- **Key Milestone:** Pioneered the core concept of **Packet Switching** rather than traditional circuit-switched telephone networks. Packet switching breaks messages into smaller blocks (packets) that travel independently through network nodes (Interface Message Processors - IMPs).
- **Legacy:** Developed early networking protocols (NCP - Network Control Program), which later evolved into the robust **TCP/IP suite** in 1983.

NSFNET (National Science Foundation Network)

- **Introduction (1986):** Established by the U.S. National Science Foundation to provide high-speed backbone connections between regional supercomputing centers.
- **Role in Expansion:** NSFNET opened network interconnectivity to university research centers and academic institutions, replacing ARPANET as the primary research backbone.
- **Transition to Commercial Internet:** In 1995, NSFNET decommissioned its backbone as commercial Network Access Points (NAPs) emerged, giving birth to today's commercial Internet.

The Modern Internet Architecture

The modern Internet is a hierarchical global federation of interconnected Autonomous Systems (AS) managed by Internet Service Providers (ISPs):

- **Tier-1 ISPs:** Global backbones (e.g., AT&T, Lumen, Telia) that route traffic across continents via high-speed submarine fiber optic cables. They peer directly with each other without paying transit fees.

- **Tier-2 ISPs:** Regional or national providers that purchase bandwidth transit from Tier-1 ISPs and peer with other Tier-2 networks to exchange local traffic.
- **Tier-3 / Access ISPs:** Local providers that deliver end-user connectivity (residential broadband, 5G cellular, corporate fiber).
- **IXPs (Internet Exchange Points):** Physical infrastructure hubs where different ISPs and Content Delivery Networks (CDNs like Cloudflare, Akamai) directly exchange traffic via peering.

■ 4. Protocol Layering & Service Abstractions

Concept of Protocol Layering

To manage the immense complexity of network hardware and software, computer networks are organized in a structured stack of **layers**, each built upon the one below it. This modularity enforces *encapsulation* and *separation of concerns*.

Key Principles of Layering

- **Abstraction:** Each layer performs specific tasks and hides internal operational details from adjacent layers.
- **Interface:** A well-defined boundary through which a lower layer offers services to the layer immediately above it.
- **Protocol:** A set of rules and conventions governing communication between equal entities (peer layers) on different machines.

Service Types: Connection-Oriented vs. Connectionless Service

Layers can offer two fundamental classes of communication services to the layer above them:

Characteristic	Connection-Oriented Service	Connectionless Service
Analogy	Telephone Call (Setup → Talk → Teardown)	Postal Mail System (Individual Envelopes)
Setup Phase	Required (Handshake establishes explicit connection state).	None (Data sent immediately without prior notice).
Packet Routing	All packets typically follow the established logical path.	Every packet (datagram) is routed independently.
Ordering & Reliability	Guaranteed ordered delivery; loss/corruption detected & retransmitted.	No guarantee of arrival order, delivery, or error recovery.
Protocols / Examples	TCP (Transport Layer), ATM, Frame Relay.	UDP, IP (Network Layer), Ethernet.

Service Primitives

A service is formally specified by a set of **primitives** (operations) available to an upper-layer process to access the service. Standard service primitives include:

1. **LISTEN** Waiting for an incoming connection request.

2. **CONNECT** Initiating a connection by sending a request to a peer.
3. **ACCEPT** Responding to a connection request, confirming establishment.
4. **DATA** Sending or receiving user data payload across an active channel.
5. **DISCONNECT** Initiating or accepting teardown of a established channel.

Relationship of Services to Protocols

It is vital to distinguish between a service and a protocol:

- **Service:** Defines *what* operations a layer provides to the layer above it via an interface. It defines abstract operations and specifies no internal implementation details.
- **Protocol:** Defines *how* those services are implemented. It is a set of rules controlling the format, syntax, timing, and semantics of packets exchanged between **peer entities** across different physical machines.

■ 5. Reference Models: OSI vs. TCP/IP

The OSI (Open Systems Interconnection) Reference Model

Developed by the International Organization for Standardization (ISO), the OSI model is an 802.11/ISO standard containing **7 distinct layers** designed as a theoretical framework for standardizing network communication.

Layer	Layer Name	Primary Function & Key Responsibilities
7	Application Layer	Provides network services directly to end-user software applications (e.g., HTTP, SMTP, FTP, DNS, SSH). Handles network access, user interfaces, and application data display.
6	Presentation Layer	Data representation, formatting, syntax transformation, data compression, and encryption/decryption (e.g., TLS/SSL, JPEG, ASCII). Ensures data sent by one system is readable by another.
5	Session Layer	Establishes, manages, synchronizes, and terminates communication sessions between applications (e.g., RPC, NetBIOS). Controls dialogue continuity (full-duplex / half-duplex, checkpointing).
4	Transport Layer	End-to-end communication, segmentation, flow control, error control, and port-based multiplexing (Data Unit: <i>Segment/Datagram</i>). Protocols: TCP, UDP.
3	Network Layer	Logical network addressing (IP addresses), packet routing across multiple networks, path determination, and traffic control (Data Unit: <i>Packet</i>). Protocols: IP, ICMP, OSPF, BGP.
2	Data Link Layer	Node-to-node framing, physical addressing (MAC addresses), error detection (CRC), and Media Access Control (Data Unit: <i>Frame</i>). Protocols: Ethernet (802.3), Wi-Fi (802.11), PPP.
1	Physical Layer	Transmission of raw, unstructured bitstreams over physical medium. Defines electrical, optical, and mechanical signaling characteristics (e.g., voltage, cables, fiber optics, connectors).

The TCP/IP Reference Model

Developed by the U.S. Department of Defense (DoD), TCP/IP is the practical architectural foundation of the modern Internet. It contains **4 functional layers**:

TCP/IP Layer	Corresponding OSI Layers	Key Protocols & Functionality
4. Application Layer	Application, Presentation, Session	HTTP, HTTPS, FTP, SSH, DNS, SMTP, DHCP. Integrates all high-level application functions, formatting, and session control.
3. Transport Layer	Transport	TCP (Transmission Control Protocol - Reliable/Connection-Oriented), UDP (User Datagram Protocol - Unreliable/Connectionless). Uses Port Numbers.
2. Internet Layer	Network	IPv4, IPv6, ICMP, ARP, IPsec. Handles logical addressing, routing, and packet packaging across disparate network technologies.
1. Link / Network Access Layer	Data Link, Physical	Ethernet, Wi-Fi, Point-to-Point Protocol (PPP). Handles hardware interfaces, MAC addresses, drivers, and physical signaling over wires/radio channels.

Comparison: OSI vs. TCP/IP Models

- **Theoretical vs. Practical:** OSI is a comprehensive conceptual model created prior to implementation. TCP/IP was designed around protocols already built and deployed in practice.
- **Layer Count:** OSI has 7 layers; TCP/IP merges the top three (Application, Presentation, Session) into a single Application layer, and the bottom two (Data Link, Physical) into a single Network Access layer.
- **Service Rigidity:** OSI strictly distinguishes between services, interfaces, and protocols. TCP/IP does not explicitly emphasize this separation, making it more flexible but less modular in theory.